Validation of Radar-defined Hydrometeor Classification via the Radar Columns and Aircraft Timeseries (RadCAT) Product

Bryce R. Rickbeil and Joseph O'Brien

ABSTRACT

The Mid-latitude Continental Convective Clouds Experiment (MC3E) was a campaign hosted by the Department of Energy's (DOE) Atmospheric Radiation Measurement (ARM) Southern Great Plains (SGP) facility from April 22 – June 6, 2011, as a collaboration between the DOE and NASA. This campaign was focused on further understanding convective storm dynamics by utilizing a collection of ground-based and aircraft-based observations to explore mesoscale convective systems (MCS). May 20, 2011, MC3E case day has had significant attention from the scientific community to test agreements between observed radar reflectivity's and in-situ aircraft observations of hydrometeor characteristics. However, the methods used for these comparisons are not openly available for the larger scientific community, limiting analysis of convective storm processes within recent and future campaigns. The Radar Columns and Aircraft Time-series (RadCAT) product was developed as an open-source method for spatiotemporally syncing radar and in-situ aircraft observations. Utilizing the Python ARM Radar Toolkit (Py-ART) and Colorado State University (CSU) Radar Tools, validation of radar defined hydrometeor classification with RadCAT is conducted. Exploration of periods identified by the C-band Scanning ARM Precipitation Radar, 2nd Generation (CSAPR2) as hail are explored, with particle images from the High Volume Precipitation Spectrometer (HVPS-3) indicating these regions are a collection of small graupel and melting large aggregates.

1. Introduction

The understanding of atmospheric processes is crucial for day-to-day life as weather impacts people and property across the world. Gaining a better understanding of these processes can further the ability to predict and mitigate dangerous and damaging situations attributed to weather, more specifically convective storm environments. To further this understanding, field campaigns are conducted using many different observation techniques to build a dataset that can provide evidence as to why such phenomena are occurring. An important observation technique that can be utilized is airborne observations where an aircraft will gather targeted in-situ data of a given phenomenon. These observations can then be used to understand the underlying microphysical dynamics that are occurring in a storm. This is important when attempting to understand phenomena, such as hail formation or excessive rainfall occurrence. Although these in-situ observations are important and useful to campaigns, they are also expensive and difficult to coordinate. This is why the use of remote sensing instruments such as weather radars are used in operational situations as they provide more spatial coverage and are also more accessible for use within field campaigns.

The Department of Energy's (DOE) Atmospheric Radiation Measurement (ARM) Southern Great Plains (SGP) facility located in north-central Oklahoma is a site dedicated to hosting field campaigns using observation methods previously mentioned. The ARM SGP site consists of an observation network spread over 160 acres of land with observations from rain gauges, laser disdrometers, lidars, and weather radars all available to users of the site. This site also provides users with the ability to create data sets that can be incorporated into model simulations of the atmosphere which can allow for further understanding of the environment around the SGP site (Sisterson et al., 2016). A field campaign hosted by the SGP site accessing convective storm dynamics was the Mid-latitude Continental Convective Cloud Experiment (hereafter referred to as MC3E). Occurring from April 22 - June 6, 2011, as a collaboration between the DOE and NASA, MC3E utilized radiosonde, C-band weather radar, laser disdrometers, in-situ microphysics aircraft, and high altitude aircraft radar observations all to gain a more complete understanding of convective storm dynamics (Jensen et al., 2016). May 20, 2011, a specific day of interest to the cloud microphysics community, had the occurrence of a squall line with a large stratiform precipitation region developing behind the storm, making for a targeted area of interest as safe aircraft observations could be made in this region. This day also had quality radar observations from the initiation of the storm to the fully developed stage in its lifecycle, along with good coordination of in-situ and high altitude aircraft observations.

Murphy et al. (2020) analyzed the May 20, 2011, MC3E case and detailed an analysis technique that paired aircraft observations with ground radar observations. Murphy et al. (2020) would extract a vertical profile from radar data which could then be compared with in-situ aircraft observations. This technique was created to gain a better understanding of the location of small ice particles that could impact commercial aircraft using a ground-based radar. Though this technique worked well within the study, there currently is no open-source version available to the general cloud microphysics community. Therefore, the development of Radar Columns and Aircraft Timeseries (RadCAT) is conducted to allow scientists the ability to create a comparison data set for any field campaign that has a microphysics aircraft and ground-based radar.

2. Methodology

The May 20, 2011, MC3E campaign day, “golden day”, was a particular day of interest for this study as good observations from the University of North Dakota (UND) Citation aircraft and

CSAPR DOE radar were collected. The convective event that ensued on May 20th was a squall line starting in the western portion of Oklahoma before going further upscale and moving northeast over the SGP ARM site. The duration of the flight was around four hours, with a dog-bone flight pattern being most common throughout the flight. The focus of the UND Citation during this flight was on the back end of the storm where a large stratiform precipitation region existed extending through the north to the south. The ground-based radar used to scan this storm was the C-band Scanning ARM Precipitation Radar (CSAPR) equipped with dual polarization capabilities. This radar can produce products such as reflectivity, radial velocity, correlation coefficient, differential reflectivity, and specific differential phase. Given this radar being in C-band, the CSAPR can measure out to ~100 km in range but does have some limitations during high precipitation events as the signal is more easily attenuated (Kollias et al., 2020). The radar data itself was processed using Corrected Moments to Antenna Coordinates (CMAC) which identifies the scatter that the radar is measuring, then quality controlling the observations and making corrections to measured anomalies as necessary (Jackson et al., 2026).

During the MC3E campaign, the UND Citation II microphysics aircraft was utilized to gather in-situ observations of the convective environments around the SGP domain. This aircraft was owned and operated by the University of North Dakota with the goal of providing in-situ cloud microphysics observations for campaigns such as MC3E which were specifically focused on cloud dynamics. During the MC3E campaign this aircraft was equipped with many microphysical instruments such as the HVPS-3, Cloud Droplet Probe, Cloud Particle Imager, Temperature Probe, Static Pressure Sensor, etc. The main instruments that were utilized with the development of RadCAT were the HVPS-3, Temperature Probe, and the onboard aircraft position instrument. The HVPS-3 or High Volume Precipitation Spectrometer is an optical array probe (OAP) designed by

Stratton Park Engineering Company (SPEC) which can measure hydrometeors between the sizes of 150-19,000 µm. This probe can also provide products such as hydrometeor images, particle size distributions, and area ratio (Glienke & Mei, 2020). The processing software for generating these products, System for OAP Data Analysis (SODA), can calculate the sizes of OAP shadow images utilizing a "best-fit circle". With knowledge of the aircraft's true air speed and the calibrated sample area of the OAP, concentrations can be calculated from the sized shadow images to form particle size distributions. Additional hydrometeor characteristics, such as particle orientation, can be determined from parameters associated with the "best-fit circle", such as area and aspect ratios (Bansemer, 2016/2026).

RadCAT, was developed using the Python libraries Python ARM Radar Toolkit (Py-ART) and Colorado State University (CSU) Radar Tools which together can generate a radar object and determine hydrometeor classification. Before radar column extraction, the RadCAT software will read through the defined radar file directory and generate a list of available radar files. With this list, radar files will then be opened in pairs of twos and valid times will be found from the last elevation scan time of the radar file, see Figure 1. This period of valid times is then compared against aircraft flight times to find overlap between the aircraft and radar which if overlap is found an appropriate radar column will be extracted for the aircraft's location at that time. The extraction function used within RadCAT is the built-in function of Py-ART *columnsect.get_field_location()*. This function takes in the latitude and longitude of the aircraft and finds the direction of the scan using the forward azimuth and distance using the haversine great-circle formula, from which it returns a vertical radar column profile at the defined location (O'Brien et al., 2025). Once this valid radar column dataset is extracted the code then uses the aircraft's position altitude to find which radar sweep the aircraft is closest to. The code will then append to the aircraft microphysics file three different radar sweeps to the location of the aircraft (e.g. at, above and below the location of the aircraft). This multi-sweep inclusion is to give the user the ability to average or do separate analysis of the surrounding radar data at the location of the aircraft. RadCAT is also able to derive reflectivity values given the particle size distribution from the HVPS-3. This calculation uses Eqs 1, 2, and 3 to find the reflectivity which uses a mass dimensional relationship method to find the spherical droplet equivalent of the given observed hydrometeors. This equation uses an alpha value of 0.00528 and a beta value of 2.1 which is used within (Heymsfield et al., 2007). There is also the inclusion of a dielectric coefficient to account for the dominant ice hydrometeors that are observed throughout this case day, the value used is .19 which is from (Chandrasekar et al., 2023). The CSU Radar Tools python library was utilized within the analysis of the May 20th Day to find the hydrometeor classification at the aircraft's location. The CSU Radar Tools function takes reflectivity, differential reflectivity, specific differential phase, correlation coefficient, radar band type, and aircraft air temperature where it then uses fuzzy logic to return the best guess hydrometeor classification. This classification is in the form of a number which corresponds to a specific hydrometeor

$$(1)\ m(D) = \alpha D^{\beta}$$
$$(2)\ Z_e = \left(\frac{6}{\pi}\right)^2 \cdot (1 \times 10^6) \sum_{i=1}^{T} m(D_i)^2 N_i$$
$$(3)\ dBZ = (.19) \cdot 10\log_{10}(Z_e)$$

Where *m(D)* is mass per particle diameter *(D)*, α and β are mass dimensional coefficients, *Ze* is equivalent radar reflectivity factor, *Ni* is number concentration at bin i, and *dBZ* is reflectivity in log scale.

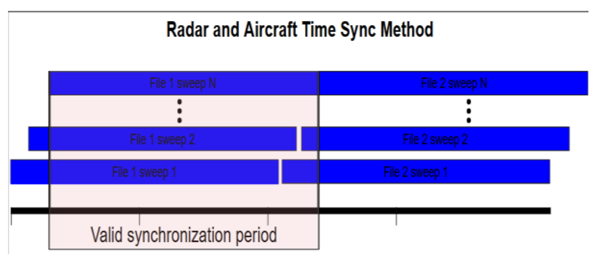

Figure 1: Illustration showing how the multiple radar files are used to synchronize the ground-based radar with the aircraft's position.

classification that is defined within the CSU Radar Tools python function. This number is also appended to the aircraft microphysics file for any overlap times between the radar files and aircraft times. This data is all contained within a netcdf file with the naming conventions being

Radar_Reflectivity, Calculated_Reflectivity, and Phase_Classification for three sweeps of radar reflectivity, the aircraft calculated reflectivity, and the CSU Radar Tools hydrometeor classification respectively.

3. Discussion

After generating the spatiotemporally synced dataset between the CSAPR radar and UND microphysics citation II aircraft, pseudo RHI plots were made to reassure that the reflectivity that was being returned from RadCAT is correct given the position of the aircraft. This plot shown in Figure 2 shows the location of the aircraft by the triangle marker along with the radar reflectivity color. Figure 2 shows the aircraft during the time of high reflectivity which is also associated with the bright banding area of the radar. This bright band is also indicative of the presence of a melting layer due to hydrometeors going through a phase change which causes higher radar return reflectivity. This region of the MCS also seemed to flag hail occurrences the most, which is most likely due to the high reflectivity values encountered in this flight leg, which are commonly associated with hail. Figure 3 shows HVPS-3 images during this flight leg, and it appears that the common hydrometeor in this region is aggregated snow and ice that also may have some melting present too. This demonstrates that CSU Radar Tools can overrepresent the presence of hail when a bright band is apparent in the radar scan which could lead to misinformed weather alerts for locations near this phenomenon. Figure 3 is a paneled display built on top of RadCAT's output data that can quickly visualize the bulk of data that RadCAT outputs. This display was used to look at not only hail events but also other hydrometeor classifications that were detected by the CSU Radar Tools library. During most of the higher altitude portions of the flight there was usually better agreement between the HVPS-3 observations and what CSU Radar Tools was identifying as the predominant hydrometeor. When comparing the radar reflectivity and aircraft calculated reflectivity, in Figure 4 the calculated value commonly underrepresents the radar measured reflectivity value. This is most likely caused by the need for a more complex hydrometeor radar reflectivity representation using a T-matrix method to find calculated reflectivity. Whereas within the scope of this study a mass dimensional relationship was used which will more closely represent the hydrometeor reflectivity to what the radar observes but will still have some errors present as it is not the most accurate representation. Using the RadCAT display software the location of the aircraft and what the radar and calculated reflectivity's were at that time were able to be analyzed. The results showed that during high altitude legs of the Citation flight the two reflectivity values agreed

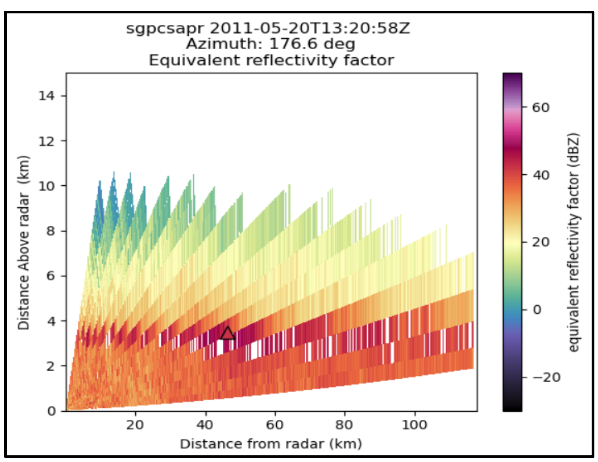

Figure 2: CSAPR pseudo RHI radar scan with aircraft location denoted by marker.

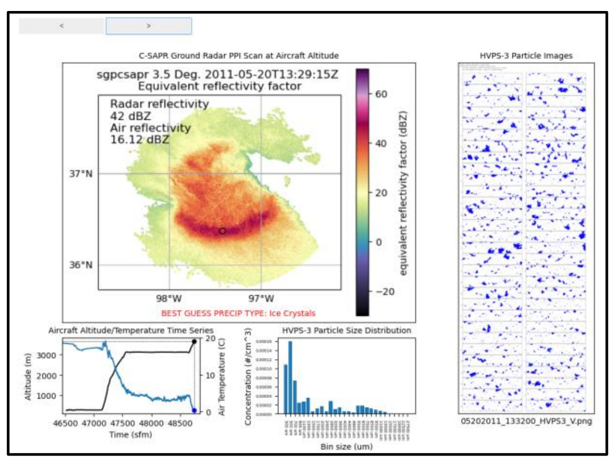

Figure 3: Interactive data display panel showing aircraft altitude and temperature time series, HVPS-3 particle size distribution and images, and CSAPR radar PPI scan.

more compared to the lower levels where a more mixed hydrometeor phase existed. Figure 4 also shows the different hydrometeor classifications throughout the dataset, where the wet snow classification period appears to have the closest agreement between the reflectivity products that are compared. This was most likely the result of the coefficients and dielectric that were used within the bulk aircraft reflectivity calculation, as these coefficients are most effective for ice hydrometeors such as wet snow. This does however cause other hydrometeor types to not be as best represented in terms of calculated radar reflectivity which can be seen in Figure 4 where rain and ice crystals have very little one-to-one agreement between the reflectivity's compared.

The RadCAT display software also showed that there were ice hydrometeors existing within a region of the melting layer that was above 0°C. This is normally unexpected as within this region the primary hydrometeor should be in liquid phase, but due to the large ice aggregate sizes and conditions that allow for sublimation to overtake evaporation these hydrometeors are able to withstand further depths of the melting layer, which aligns with what was found in (Skow, et al. 2022). More above this region there also appeared to be long aggregates present indicated by Figure 5 where there were ice aggregate hydrometeors around 2-3cm in length. Given this

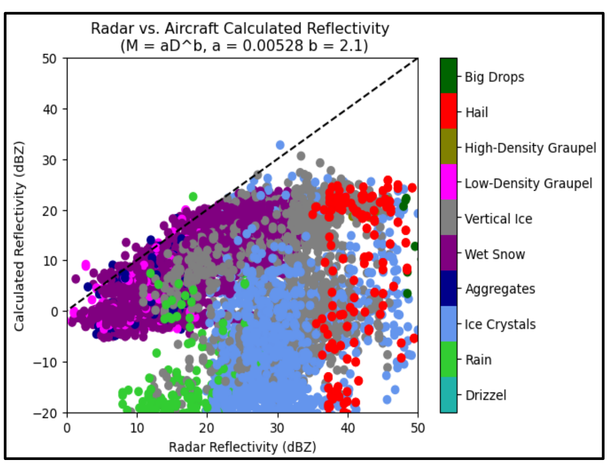

Figure 4: Radar reflectivity and aircraft calculated reflectivity comparison with hydrometeor phase classification.

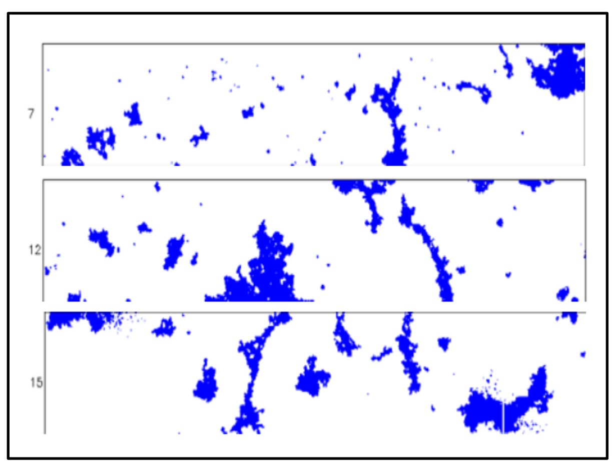

Figure 5: HVPS-3 images during ascent through the melting layer from the 13:32:00 UTC photo collection, these being the 7, 12, and 15 second frames from the collection.

observation it is possible that a sufficiently strong electric charge layer was present that caused larger aggregates to form given findings in (Nairy et al., 2026).

4. Conclusion

The Midlatitude Convective Continental Clouds Experiment was conducted at the Atmospheric Radiation Measurement Southern Great Plains site to collect data to better understand convective storm dynamics. With this collected data set a method of syncing ground-based radar data with airborne microphysics aircraft data was made to allow for comparisons between these two observation methods. RadCAT built on Py-ART and CSU Radar Tools can sync these two datasets and generate a comparison product along with hydrometeor classification throughout the duration of the flight to then compare this dataset with HVPS-3 observations on board the University of North Dakota Citation II aircraft. With this spatiotemporal synchronization of the aircraft observations with the radar observations products such as reflectivity, hydrometeor classification, etc. can be compared between the two methods which can contribute to a better understanding of different regions of a convective cloud given radar attributes and aircraft observations.

Specifically, looking at areas where hail was classified by CSU Radar Tools, the UND Citation observed larger aggregates of ice with some water covering the outer layer present too. This revealed that within the melting layer CSU Radar Tools was more likely to classify the hydrometeors as hail, which could have been caused by the higher reflectivity values as a result of this being the bright band region of the radar scan. Though no hail was observed, larger almost chained ice aggregates did seem to be present which could indicate that an electric field was present within this region of the storm. RadCAT would also be able to perform this same kind of analysis for other CSU Radar Tools defined hydrometeor types which could even further reveal other important characteristics of not only the May 20th case but other case days and even other campaigns with similar observation methods. With a larger dataset created by RadCAT further analysis using Machine Learning models could also be of interest as further trends could be identified within the synchronized data which could further the field of cloud microphysics and allow future investigations into hydrometeor classification aloft.

5. Acknowledgments

I would like to thank the Department of Energy Science Undergraduate Laboratory Internship for supporting me this summer as an intern and giving me the opportunity to work on this project. I would like to thank Argonne National Laboratory for hosting me this summer. I would also like to thank the National Aeronautics and Space Administration, the University of North Dakota, and the Atmospheric Radiation Measurement facility for providing all the raw and processed data that was used throughout this study. Then finally, I would like to thank Joseph O’Brien for having me as an intern and guiding me through the development of RadCAT along with providing great future resources for my career to come. This work was supported in part by the U.S. Department of Energy, Office of Science, Office of Workforce Development for Teachers and Scientists (WDTS) under the Science Undergraduate Laboratory Internships (SULI) Program.

References

Bansemer, A. (2026). *Abansemer/soda2* [IDL]. https://github.com/abansemer/soda2 (Original work published 2016)

Glienke, S., & Mei, F. (2020). *High-Volume Precipitation Spectrometer (HVPS) Instrument Handbook* (DOE/SC-ARM-TR--239, 1597643; p. DOE/SC-ARM-TR--239, 1597643). https://doi.org/10.2172/1597643

Heymsfield, A. J., Bansemer, A., & Twohy, C. H. (2007). Refinements to Ice Particle Mass Dimensional and Terminal Velocity Relationships for Ice Clouds. Part I: Temperature Dependence. *Journal of the Atmospheric Sciences*, *64*(4), 1047–1067. https://doi.org/10.1175/JAS3890.1

Jackson, R., O'Brien, J., Grover, M., Sherman, Z., Collis, S., Raut, B. A., Tuftedal, M., Theisen, A., Feldman, D., Rudisill, W., & Chandrasekar, V. (2026). Surface Quantitative Precipitation Estimates (SQUIRE) of Snow Water Equivalent from the Surface Atmospheric Integrated Field Laboratory. *Journal of Atmospheric and Oceanic Technology*, *43*(1), 77–93. https://doi.org/10.1175/JTECH-D-25-0023.1

Jensen, M., Kollias, P., & Giangrande, S. (2014). *The Mid-latitude Continental Convective Clouds (MC3E) Experiment Final Campaign Report* (DOE/SC-ARM--14-012, 1192141; p. DOE/SC-ARM--14-012, 1192141). https://doi.org/10.2172/1192141

Jensen, M. P., Petersen, W. A., Bansemer, A., Bharadwaj, N., Carey, L. D., Cecil, D. J., Collis, S. M., Genio, A. D. D., Dolan, B., Gerlach, J., Giangrande, S. E., Heymsfield, A., Heymsfield, G., Kollias, P., Lang, T. J., Nesbitt, S. W., Neumann, A., Poellot, M., Rutledge, S. A., … Zipser, E. J. (2016). The Midlatitude Continental Convective

Clouds Experiment (MC3E). *Bulletin of the American Meteorological Society*, *97*(9), 1667–1686. https://doi.org/10.1175/BAMS-D-14-00228.1

Kollias, P., Bharadwaj, N., Clothiaux, E. E., Lamer, K., Oue, M., Hardin, J., Isom, B., Lindenmaier, I., Matthews, A., Luke, E. P., Giangrande, S. E., Johnson, K., Collis, S., Comstock, J., & Mather, J. H. (2020). The ARM Radar Network: At the Leading Edge of Cloud and Precipitation Observations. *Bulletin of the American Meteorological Society*, *101*(5), E588–E607. https://doi.org/10.1175/BAMS-D-18-0288.1

Murphy, A. M., Ryzhkov, A., & Zhang, P. (2020). Columnar Vertical Profile (CVP) Methodology for Validating Polarimetric Radar Retrievals in Ice Using In Situ Aircraft Measurements. *Journal of Atmospheric and Oceanic Technology*, *37*(9), 1623–1642. https://doi.org/10.1175/JTECH-D-20-0011.1

Nairy, C. M., Delene, D. J., Finlon, J. A., Yorks, J. E., Järvinen, E., Schnaiter, M., Walker McLinden, M. L., Heymsfield, A. J., Detwiler, A. G., & McMurdie, L. A. (2026). In Situ Observations of Ice Crystal Chain Aggregates in Winter Storms. *Geophysical Research Letters*, *53*(1), e2025GL118365. https://doi.org/10.1029/2025GL118365

O'Brien, J., Grover, M., Jackson, R., Sherman, Z., Collis, S., & Theisen, A. (2025). *Extracted Radar Columns and In Situ Sensors (RadCLss) Value-Added Product Report*.

Sisterson, D. L., Peppler, R. A., Cress, T. S., Lamb, P. J., & Turner, D. D. (2016). The ARM Southern Great Plains (SGP) Site. *Meteorological Monographs*, *57*(1), 6.1-6.14. https://doi.org/10.1175/AMSMONOGRAPHS-D-16-0004.1

Skow, A., Poellot, M., Delene, D., Askelson, M., North, K., & Oue, M. (2022). *A Multi-Platform In Situ Kinematic and Microphysical Analysis of a Hybrid Parallel–Trailing Stratiform Mesoscale Convective System*. https://doi.org/10.1175/MWR-D-21-0009.1